Chapter 03. Optimal Risky Portfolio

Shushu Liao

Investment I

Peking University Business School UK Campus

Chapter Outline

- Revision of Portfolio Math
- Diversification
- Optimal risky portfolio
- The Markowitz portfolio optimization model
- Excel Practice

Risky Asset

- Previously, we understand how to allocate capital between a risky asset and the risk-free asset, we need to show that how to create a single optimal risky portfolio.
- Let us start with two risky securities.
- How should you combine *two* risky securities in your portfolio?
  - To answer this question, we will plot the possible set of expected returns and standard deviations for different combinations of two risky assets
  - We will define the *Minimum Variance Frontier*, or the set of portfolios with the lowest variance for a given expected return

Let us first review the portfolio maths.

Portfolios of Two Risky Assets: Return

- Portfolio return: $r_p = w_A r_A + w_B r_B$
  - w_A = Security A weight
  - r_A = Security A return
  - w_B = Security B weight
  - r_B = Security B return

  $E(r_p) = w_A E(r_A) + w_B E(r_B)$

Portfolios of Two Risky Assets: Variance

Portfolio variance:

- $\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \sigma_{AB}$

σ_A^2= Security A variance σ_B^2 = Security B variance

σ_{AB} = Covariance of returns for Security A and B (sometimes it is written as $\mathrm{Cov}(\mathrm{r_A}, \mathrm{r_B})$)

Portfolios of Two Risky Assets: Correlation

- Covariance of returns on security A and B:

$$\sigma_{AB} = \rho_{AB}\sigma_A\sigma_B$$

  - $\rho_{A,B}$ = Correlation coefficient of returns
  - σ_A = Standard deviation of security A
  - σ_B = Standard deviation of security B

Portfolios of *n* Risky Assets

- Expected Return: $\sum_{i=1}^{n} w_i E(R_i)$
- Portfolio Variance: $\sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j Cov(R_i, R_j)$

Portfolios: Matrix Format

- Denote the *nx1* weighting vector as w
- Denote the *nx1* expected return vector as E
- Denote the *nxn* variance-covariance matrix as Σ
- Portfolio return $\mathrm{w}'\mathrm{E}$
- Portfolio return $w'\Sigma w$

Chapter Outline

- Revision of Portfolio Math
- **Diversification**
- Optimal risky portfolio
- The Markowitz portfolio optimization model
- Excel Practice

Diversification

- If you only invest in Digital, Digital's success in research and development and personnel changes affect your investment return Digital
- However, this does not noticeably affect other firms in the economy
- Consider a diversification strategy, in which you include additional securities in your portfolio. For example, place half your funds in *ExxonMobil* and half in *Digital*.
- What should happen to portfolio risk?
- For example, when oil prices fall, hurting ExxonMobil, computer prices might rise, helping Digital. The two effects are offsetting and stabilize portfolio return.

Efficiency of Diversification I

- Consider a naïve diversification strategy: building an **equally weighted** portfolios of several securities.
- An *efficient* diversification is a risky portfolio that provide the **lowest** possible risk for any given level of expected return.

Efficiency of Diversification II

- The efficiency of diversification depends on the **correlation** between the returns of the assets in the portfolio
  - Covariance and the correlation coefficient provide a measure of the way returns of two assets move together (*covary*)
- Why?
  - Let us look at how to prove it!

Power of Diversification

- The Power of Diversification
  - Portfolio variance: $$\sigma_p^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j Cov(R_i, R_j)$$
  - *For an equally weighted* portfolio, $w_i = \frac{1}{n}$
  - We define the average variance and average covariance of the securities as:
  - $\overline{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}\sigma_i^2$
  - $\overline{cov} = \frac{1}{n(n-1)}\sum_{i=1}^{n}\sum_{j=1}^{n} Cov(R_i, R_j)$

Power of Diversification

- We can then express portfolio variance as
- $\sigma_p^2 = \frac{1}{n}\overline{\sigma^2} + \frac{n-1}{n}\overline{cov}$
  - When n becomes very large due to diversification, the first term is driven to zero
  - Portfolio variance can be driven to zero if the average covariance is zero
  - The irreducible risk of a diversified portfolio depends on the **covariance** of the returns, which is a function of the **systematic factors** in the economy

Power of Diversification

On average, portfolio risk does fall with diversification, but the power of diversification to reduce risk is limited by **systematic** or common sources of risk.

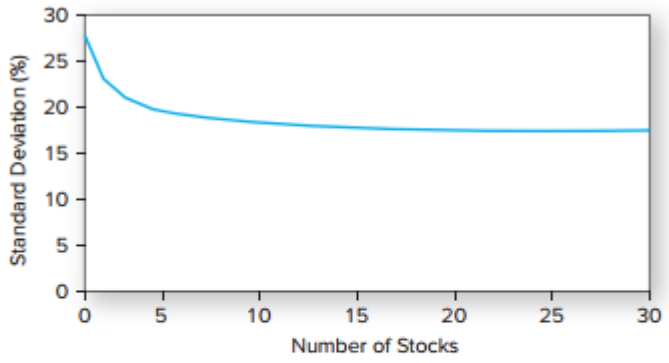
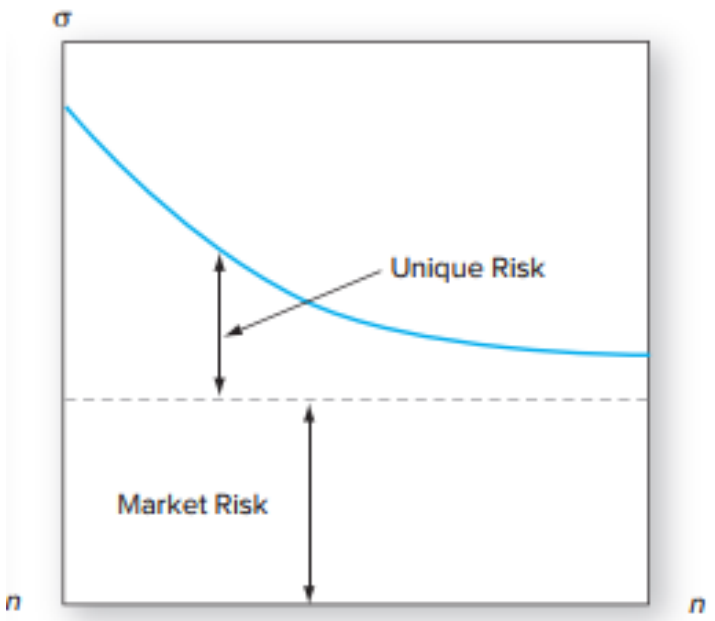


Figure 7.2 Portfolio diversification. The average annualized standard deviation of returns of portfolios composed of only one stock selected at random from the NYSE was 27.8% in the decade ending 2017. Portfolio risk fell rapidly as the number of stocks included in the portfolio increased. In the limit, portfolio risk could be reduced to only 17.4%.

Source: Authors' calculations. This is an update of a figure that first appeared in Meir Statman, "How Many Stocks Make a Diversified Portfolio?" *Journal of Financial and Quantitative Analysis* 22 (September 1987).

Market vs. Unique Risk

- Portfolio standard deviation falls as the number of securities increases, but it cannot be reduced to zero.
- The risk that remains even after extensive diversification is called **market risk** that is attributable to marketwide risk sources.
  - Such risk is also called **systematic risk,** or **nondiversifiable risk.**
- In contrast, the risk that *can* be eliminated by diversification is called **unique risk, firm-specific risk, nonsystematic risk,** or **diversifiable risk.**



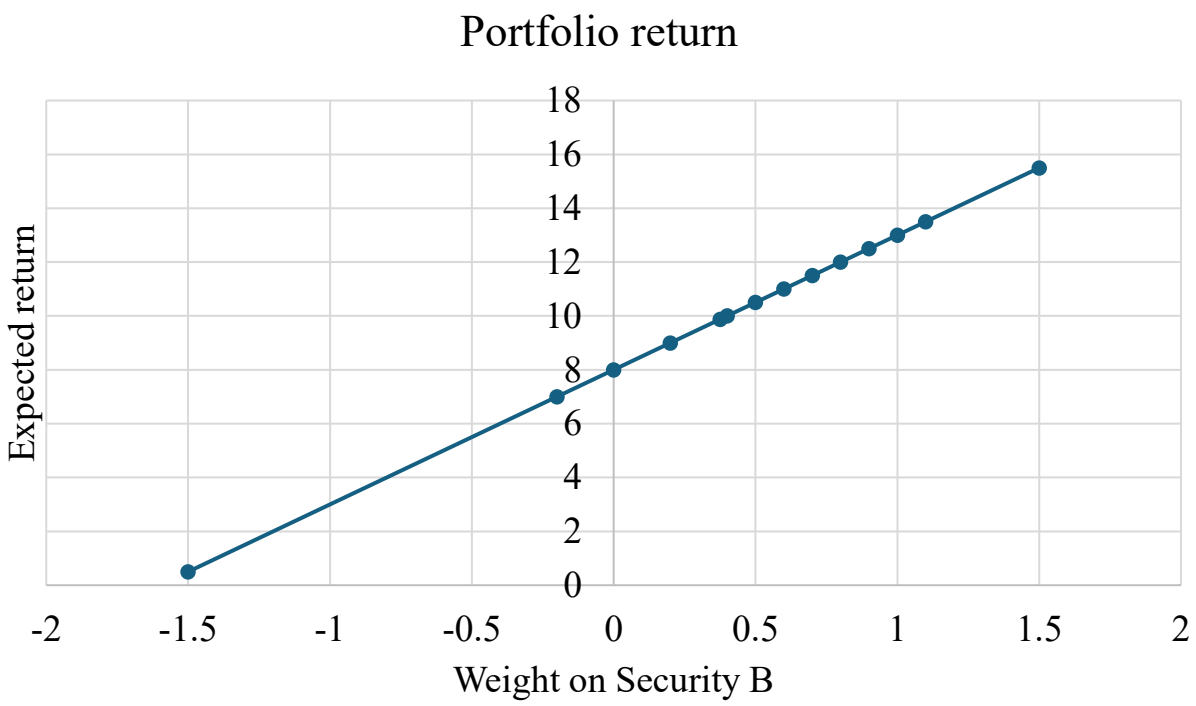
Portfolios of Two Risky Assets: Correlation

- Consider the following information for security A and B:

	$E(r)$	σ
Security A	8%	12%
Security B	13%	20%

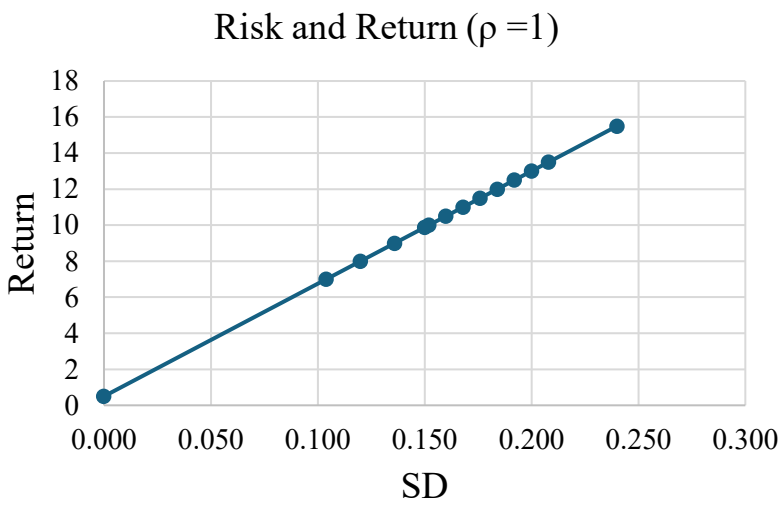
  - If ρ = 1.0, the securities are perfectly positively correlated
  - If $\rho = 0$, the securities are not correlated
  - If $\rho = -1.0$, the securities are perfectly negatively correlated

Portfolio Expected Return



- $E(r_p) = w \cdot E(r_B) + (1 - w) \cdot E(r_A)$

Portfolio Standard Deviation $\rho_{AB} = 1$



- When the two assets are perfectly correlated, it is possible to construct a "synthetic" **riskless** asset, or a **perfect hedge**
  - Intuition: When A is up, B is also up, you can build a riskless portfolio by shorting security B and buying security A.
- There is no reduction (diversification) of the risk since its standard deviation is equal to the weighted average of the component standard deviation.

Portfolio Standard Deviation $\rho_{AB} = 1$

How to find the perfect hedge?

When $\rho_{AB} = 1$

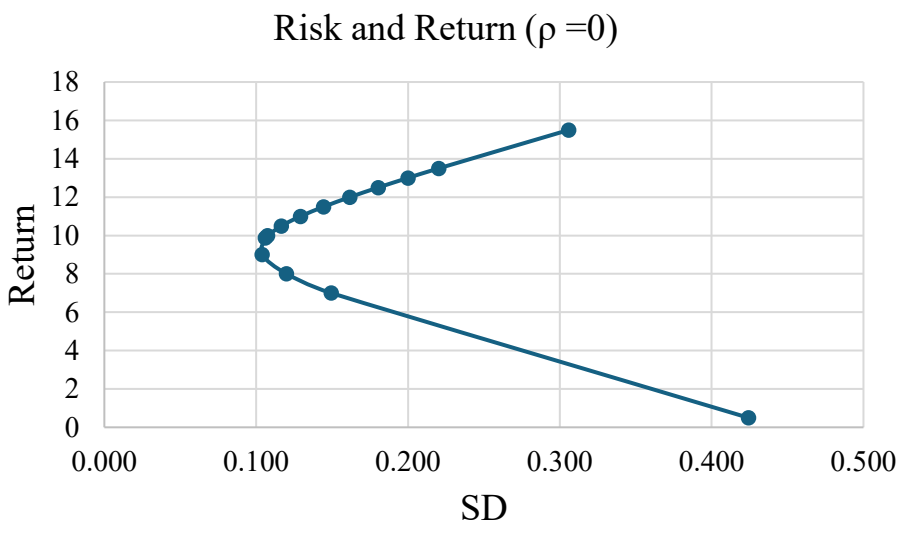
$$\sigma_p = w_A\sigma_A + w_B\sigma_B$$

- Set $\sigma_p = 0$, $w_A = -\frac{w_B\sigma_B}{\sigma_A}$
- $w_A + w_B = 1$

By substitution, $w_B = \frac{1}{1-\frac{\sigma_B}{\sigma_A}} = \frac{1}{1-20/12} = -1.5$

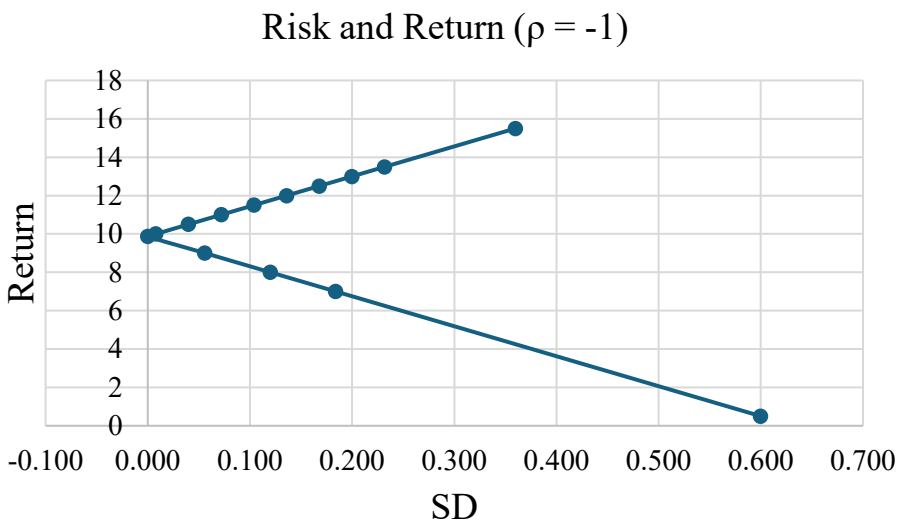
The perfect hedge has the expected return of $-1.5 \times 13 + 2.5 \times 8 = 0.5\%$

Portfolio Standard Deviation $\rho_{AB} = 0$



- When the two assets are not correlated
- Because the portfolio's expected return is the weighted average of its component expected returns, whereas its standard deviation is less than the weighted average of the component standard deviation
  - There is some reduction (diversification) of the risk

Portfolio Standard Deviation ρ_{AB} = -1



- When the two assets are perfectly negatively correlated, it is possible to construct a "synthetic" riskless asset, or a perfect hedge
  - Intuition: When A is up, B is also down, you can build a riskless portfolio by buying both security B and A.
- Diversification is most efficient.

Portfolio Standard Deviation ρ_{AB} = -1

How to find the perfect hedge?

When $\rho_{AB} = -1$

$$\sigma_p = |w_A\sigma_A - w_B\sigma_B|$$

Think for 5 minutes.

Portfolio Standard Deviation ρ_{AB} = -1

How to find the perfect hedge?

When $\rho_{AB} = -1$

$$\sigma_p = w_A\sigma_A - w_B\sigma_B$$

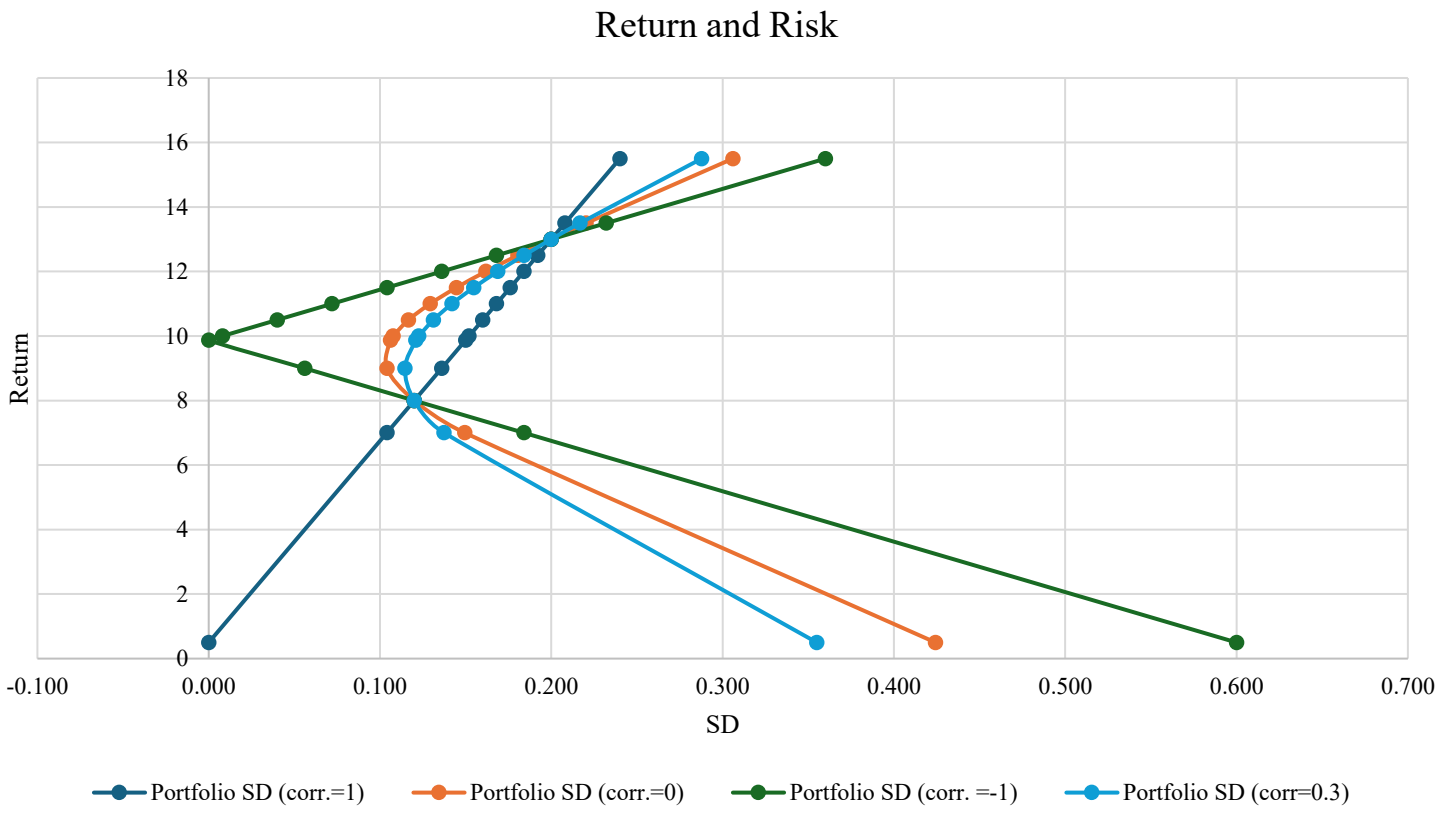
- Set $\sigma_p = 0$, $w_A = \frac{w_B\sigma_B}{\sigma_A}$
- $w_A + w_B = 1$

By substitution, $w_B = \frac{1}{1+\frac{\sigma_B}{\sigma_A}} = \frac{1}{1+20/12} = 0.375$

The perfect hedge has the expected return of $0.375 \times 13 + 0.625 \times 8 = 9.875\%$

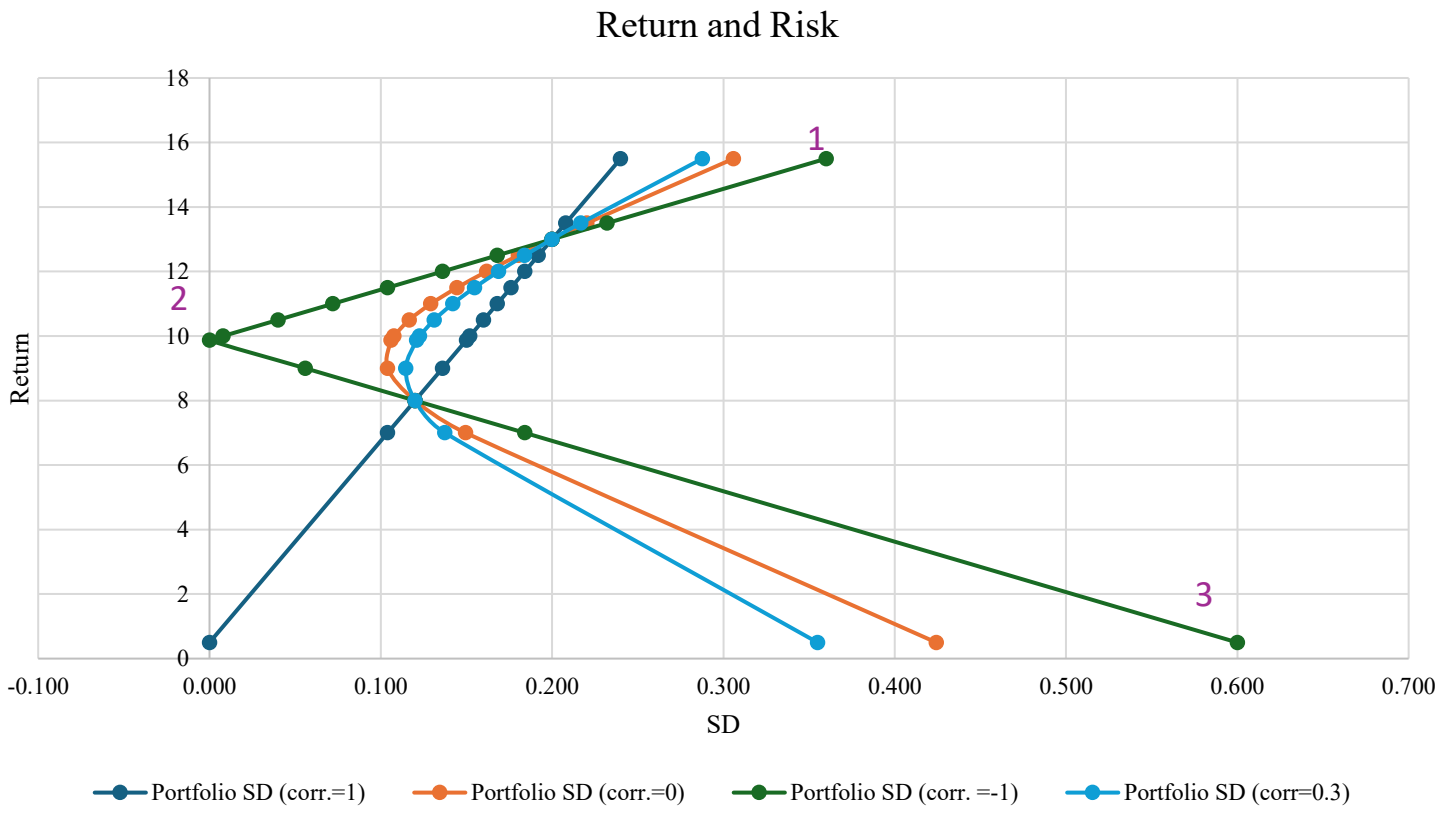
Efficiency of Diversification I

- For <u>an equally weighted portfolio</u>, the expected return is $\frac{8\%+13\%}{2} = 10.5\%$
- For 10.5%, the case of $\rho = -1$ generates the lowest SD.



Efficiency of Diversification II

- For the case of $\rho = -1$
- From 1 to 2 =>
  - Portfolio risk first falls with the initial diversification
- From 2 to 3 =>
  - Portfolio risk rises again as the portfolio becomes heavily concentrated in one asset, hence is undiversified.



Minimum Variance Portfolio

- The *minimum variance portfolio* is the portfolio composed of the risky assets that has the smallest standard deviation; the portfolio with least risk

Solution to Minimum Variance Portfolio

Objective function is

$$\min_{w_B} w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \sigma_{AB}$$

Subject to $w_A + w_B = 1$

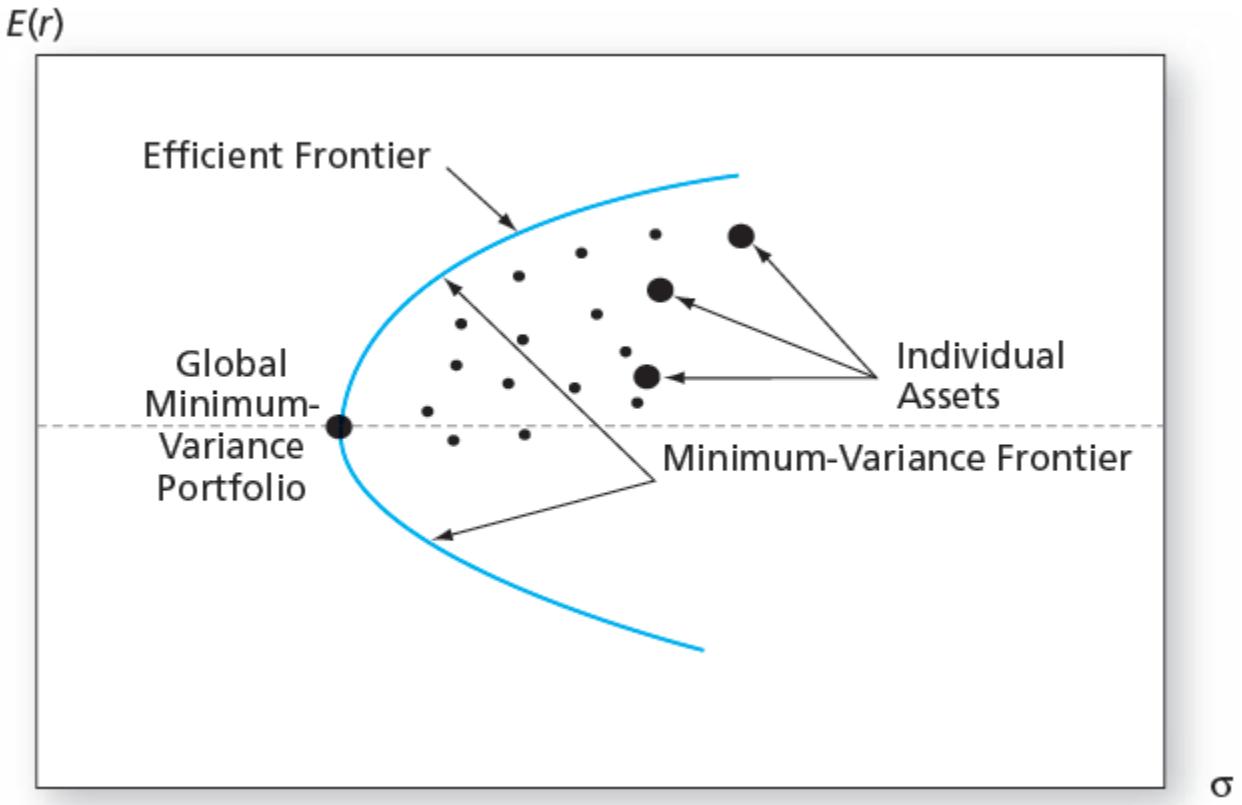
10 minutes:

Prove that the solution to minimum variance portfolio is

$$w_B = \frac{\sigma_A^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}}$$

Efficient Frontier

- For any portfolio on the lower portion of the minimum-variance frontier, there is a portfolio with the same risk and a greater expected return positioned directly above it.
- The portion of the frontier that lies above the global minimum-variance portfolio, therefore, is called the **efficient frontier of risky assets.**



Involving Risk-free Asset

- The second part of the optimization plan involves the **risk-free asset**.
- As before, we search for the capital allocation line with the highest Sharpe ratio

Asset Allocation with Two Risky Securities

- We previously derived the CAL with *one* risky portfolio *P* and a risk-free asset.
- Let's assume that risk-free asset $r_f = 5\%$, we can now trade in the *two* risky: security A and B.
- The correlation between security A and B is $\rho_{AB} = 0.3$
- Consider two risky portfolios:
  1. Minimum-variance portfolio
  2. A portfolio that invests 70% in A and 30% in B

Risky Portfolio 1

- $w_B = \frac{\sigma_A^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}} = \frac{12\%^2 - 0.3 \times 12\% \times 20\%}{12\%^2 + 20\%^2 - 2 \times 0.3 \times 12\% \times 20\%} = 18\%$
- $w_A = 82\%$
- $E(r_p) = 18\% \times 13\% + 82\% \times 8\% = 8.9\%$
- $\sigma_p = 11.45\%$
- Sharpe Ratio = (8.9 – 5) /11.45= 0.34

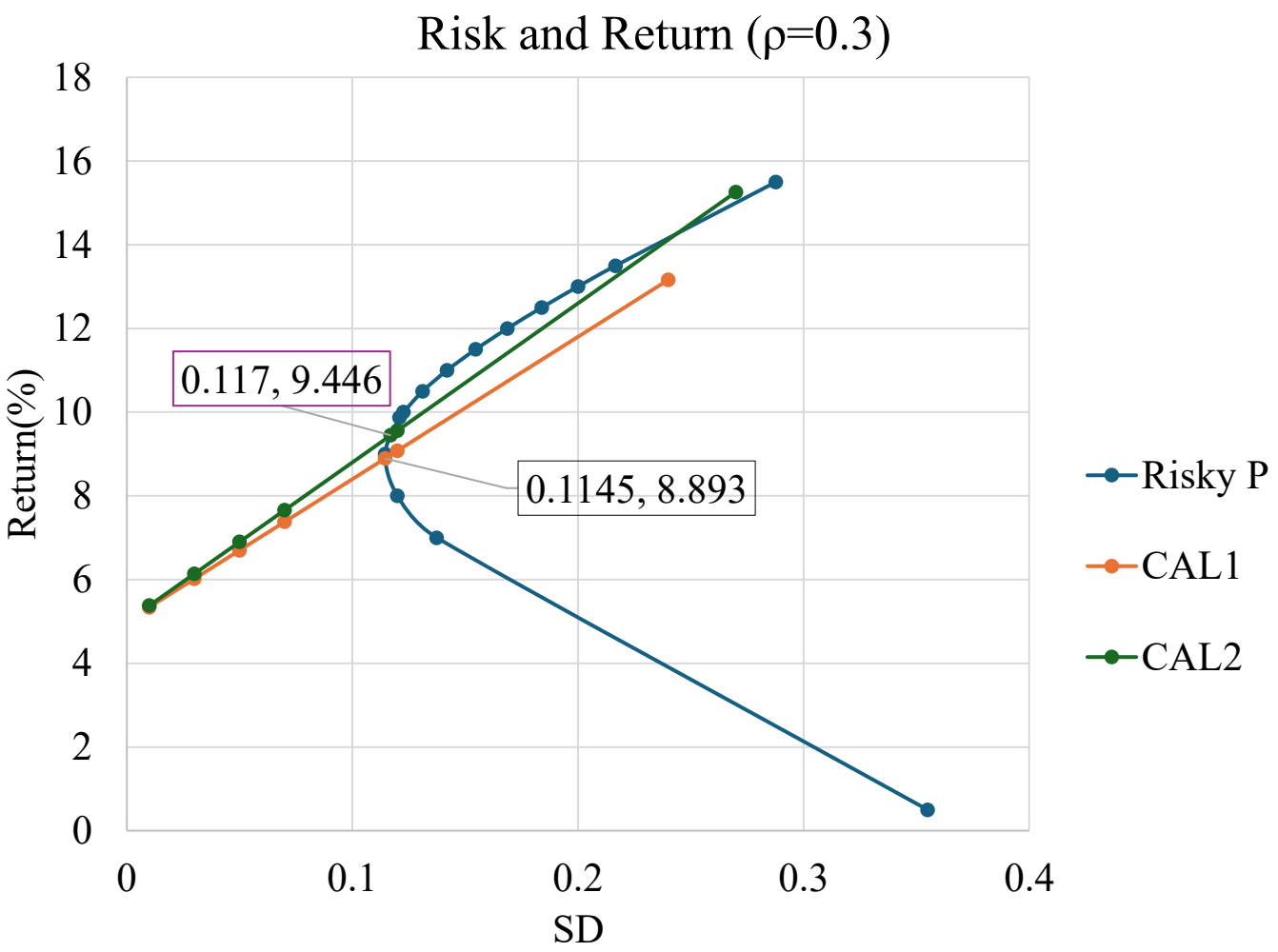
Risky Portfolio 2

- $w_B = 30\%$
- $w_A = 70\%$
- $E(r_p) = 30\% \times 13\% + 70\% \times 8\% = 9.5\%$
- $\sigma_p = 11.7\%$
- Sharpe Ratio = (9.5 – 5) /11.7= 0.38
- which is higher than the Sharpe ratio provided by the minimum-variance portfolio.
- Hence, portfolio *2* dominates *1*

The Opportunity Set of the Risky Portfolio and Two Feasible CALs

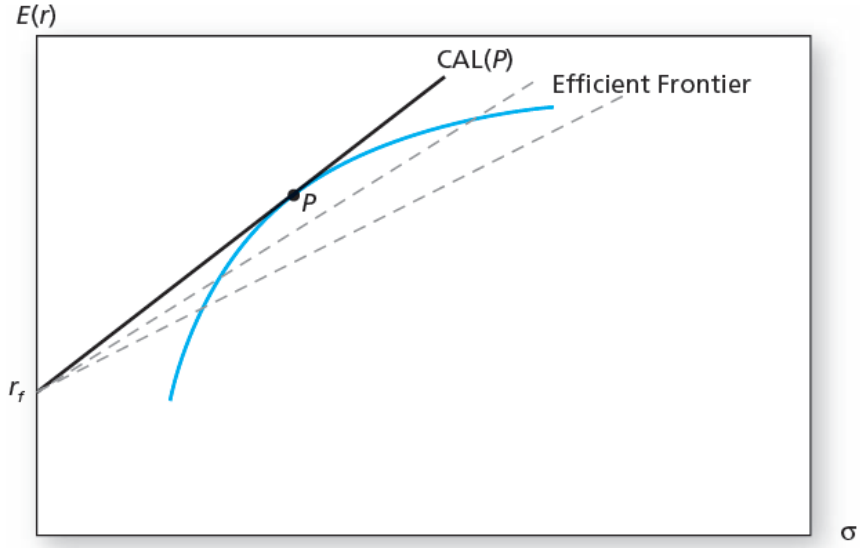
CAL*2* provides a higher expected return than CAL*1* for any level of standard deviation.

- Why should we stop at CAL2 when we can continue with a higher Sharpe ratio?



Portfolio with Highest Sharpe Ratio

- Search for the CAL with the highest reward-to-variability (Sharpe) ratio
- Everyone invests in *P*, regardless of their degree of risk aversion
  - More risk averse investors put more in the risk-free asset
  - Less risk averse investors put more in *P*



Optimal Risky Portfolio I

- Maximize the slope of the CAL for any possible portfolio, *P*
- The objective function is the slope:

$$S_p = \frac{E(r_p) - r_f}{\sigma_p}$$

Optimal Risky Portfolio II

We solve an optimization problem written as

$$\max_{w_B} \frac{E(r_p)-r_f}{\sigma_p}$$

Subject to $w_A + w_B = 1$

One can prove that the solution is (note that R denotes excess return)

$$w_A = \frac{E(R_A)\sigma_B^2 - E(R_B)\sigma_{AB}}{E(R_A)\sigma_B^2 + E(R_B)\sigma_A^2 - [E(R_A) + E(R_B)]\sigma_{AB}}$$

Optimal Risky Portfolio III

Back to the previous example. Answer

1) What is the optimal weight for security A?
2) What is the portfolio's expected return and standard deviation?
3) What is the portfolio's Sharpe ratio?

Answer

The optimal weight for A is

$$w_A = \frac{0.03 \times 0.2^2 - 0.08 \times 0.3 \times 0.2 \times 0.12}{0.03 \times 0.2^2 + 0.08 \times 0.12^2 - [0.08 + 0.03] \times 0.3 \times 0.2 \times 0.12} = 0.4$$

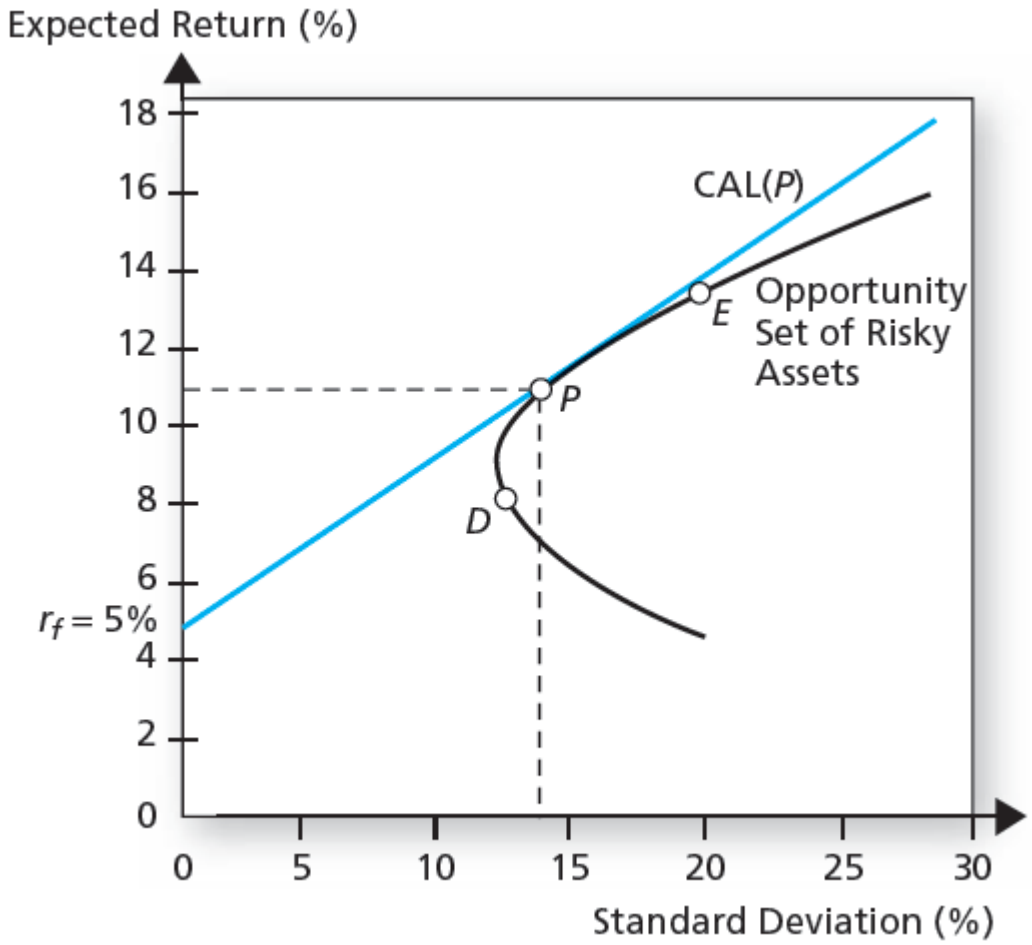
$$w_B = 0.6$$

$$E(r_p) = 0.4 \times 8 + 0.6 \times 13 = 11\%$$

$$\sigma_p = (0.4^2 0.12^2 + 0.6^2 0.2^2 + 2 \cdot 0.3 \cdot 0.4 \cdot 0.6 \cdot 0.12 \cdot 0.2)^{0.5} = 14.2\%$$

Sharpe ratio = (11-5)/14.2=0.42

Optimal Risky Portfolio III



The optimal risky portfolio *P* that maximizes the Sharpe ratio is the tangency point of a line from *5%* to the efficient frontier.

Identify Complete Portfolio *C*

- Knowing that $E(r_p) = 11\% \quad \sigma_p = 14.2\%$
- What is the investment strategy for a client with the risk-aversion score of 4. Think about:
  - What is the weight on Port. *R* *(*denoted as *y*)*?
  - What is the weight on Security A and B?
  - What is Portfolio C's expected return and standard deviation?

Answer

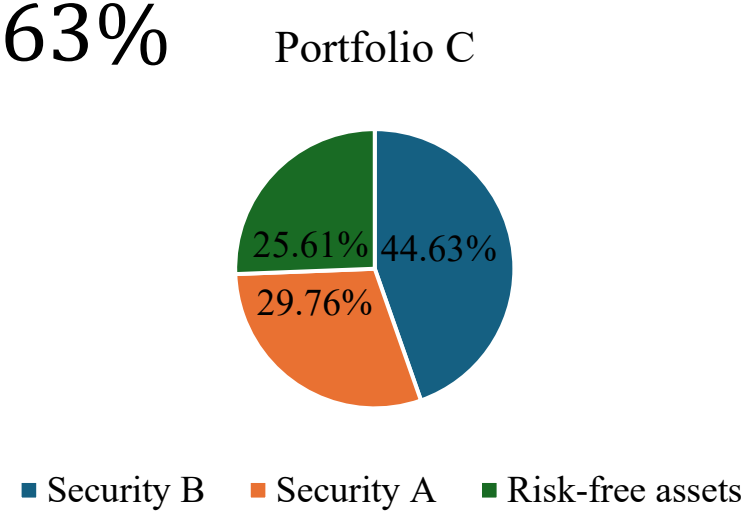
What is the weight on Port. *R* (denoted as *y**)?

$$y^* = \frac{0.11 - 0.05}{4 \times 0.142^2} = 74.39\%$$

What is the weight on Security A and B?

$$yw_A = 74.39\% \times 40\% = 29.76\%$$

$$yw_B = 74.39\% \times 60\% = 44.63\%$$



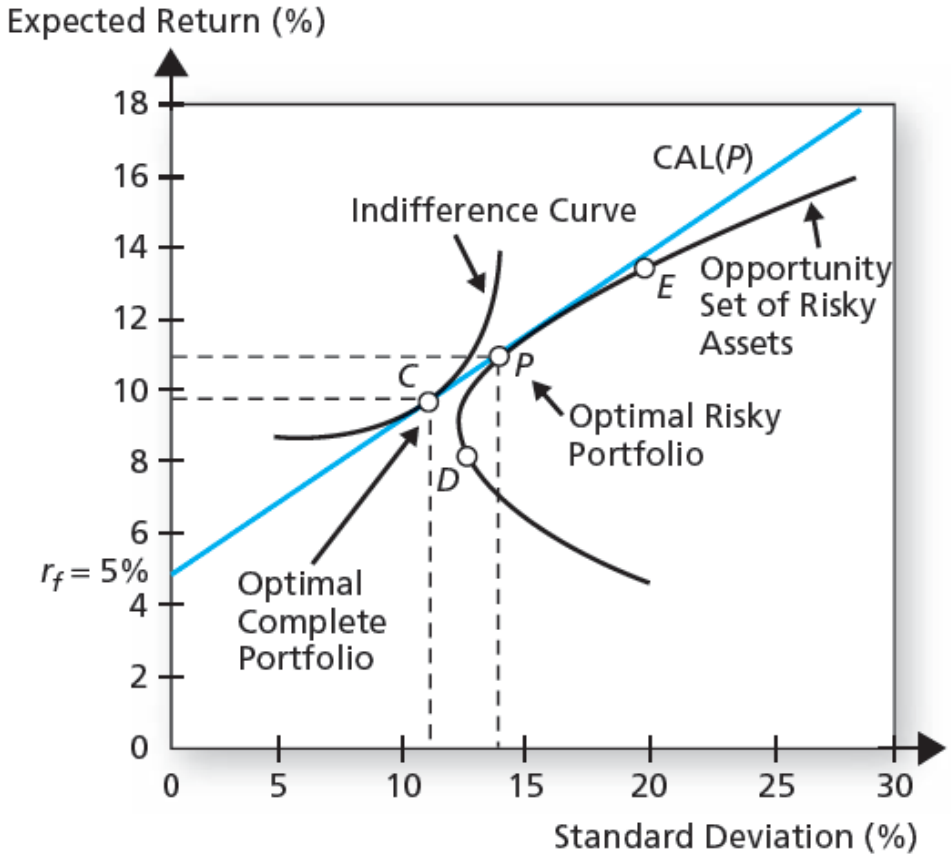
Answer

What is Portfolio C's expected return and standard deviation?

$$5\% + 74.69\%(11\% - 5\%) = 9.48\%$$

$$74.69\% \times 14.2\% = 10.606\%$$

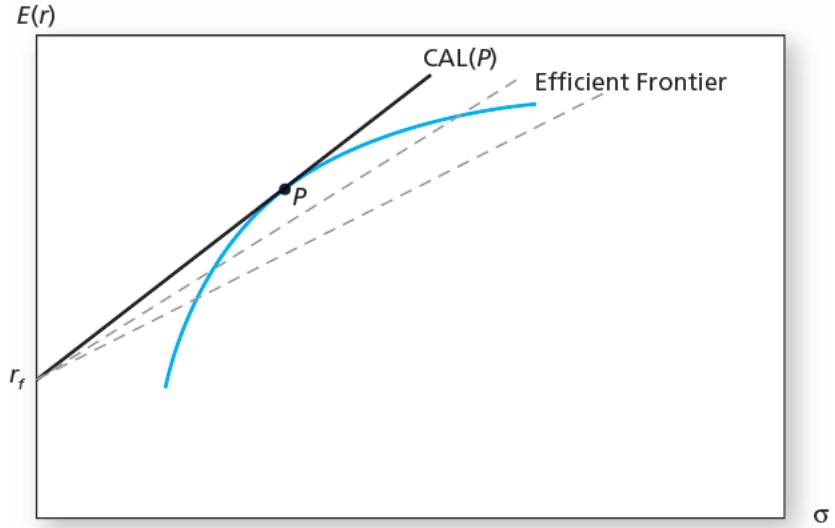
Identify Complete Portfolio C



- Given the manager's input list, the best risky portfolio *P* is the same for all clients, regardless of risk aversion.
- However, the second task, capital allocation, depends on personal preference (risk aversion).

Markowitz Portfolio Optimization

1) Security selection
   - All portfolios that lie on the minimum-variance frontier from the global minimum-variance portfolio and upward provide the best risk-return combinations (efficient frontier)
   - Search for the CAL with the highest reward-to-variability ratio (Portfolio *P*)



Markowitz Portfolio Optimization

2) Allocation of complete portfolio *C*

- Everyone invests in *P*, regardless of their degree of risk aversion
  - More risk averse investors put more in the risk-free asset
  - Less risk averse investors put more in *P*

Markowitz Portfolio Optimization Model

- Capital Allocation and the **Separation Property**
  - Portfolio choice problem may be separated into two independent tasks
    - Determination of the optimal risky portfolio is purely technical
    - Allocation of the complete portfolio to risk-free versus the risky portfolio depends on personal preference

Excel Practice

- Portfolio_pratice
- Assume the market consists of three stocks: AES, BSX, META construct the best portfolio assuming you have access to a risk-free asset with the return of 2%.
  - Step one: compute the stock returns in “PortRisky” using the adjusted stock price in “Prices”, and the annualized return (×252) and annualized variance-covariance matrix.
  - Step two (portfolio risk and return): Use portfolio matrix to compute portfolio risk and return using the sample weight.
  - Step three (the Markowitz mean-variance): Use excel solver to determine the optimal weight for the risky portfolio (portfolio *P*).

Excel Practice

- Portfolio pratice
- Assume the market consists of three stocks: AES, BSX, META
- Assume that one can use sample average return to proxy for expected return
- construct the best portfolio assuming you have access to a risk-free asset with the return of 2%.
  - Step four: In Sheet "CAL", compute the optimal weight of the portfolio *P (*denoted as *y*)* for each risk aversion score. Also, compute the weight on each stock after multiplying the weight in Step three by *y**.
  - Step five: We now repeat the Markowitz mean-variance analysis with the inclusion of a risk-free asset having a return of 2%. Then compare the result in this step to step four.

Summary

- Correlations are extremely important for determining portfolio characteristics
  - In large portfolio, only correlation matters.
- Important lessons from Mean-Variance (MV) analysis
  - You should hold the same MVE portfolio of risky assets no matter what your tolerance for risk
  - Caveat: You have to include all assets in the analysis, including human capital, real estate, etc.
  - Caveat: Markowitz's theory tells us nothing about the expected returns, variances, and covariances required to implement the theory